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ORIGINAL ARTICLE

Machining performance of PCD by ultrasound-assisted mixed-powder EDM with Cu-Ni electrode

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Abstract
Polycrystalline diamond (PCD), as a composite of diamond particles and cobalt binder sintered under high temperature and pressure, is widely applied in many fields. EDM is recognized as a suitable machining method for PCD, although it commonly suffers from poor surface quality of the machined workpiece. This study aimed to investigate the machining of PCD with Cu-Ni electrodes under the ultrasound-assisted mixed-graphene powder EDM process. Specifically, the cobalt removal mechanism and the effect on the surface and properties of the machined workpieces were analyzed. Then, the effect of cobalt removal was verified by the friction wear experiments to explore the tribological properties of the workpieces. The results suggest that the cobalt content of the Cu-Ni electrode-machined workpiece was 63% and 40% lower than that of the untreated workpiece and the Cu electrode-machined workpiece, respectively. Compared with the Cu electrode, the Cu-Ni electrode-machined surface exhibited a porous structure composed of numerous pores, attributed to the presence of ultrasonic and Ni elements. Under lubricated friction conditions, the friction coefficient of the Cu-Ni electrode-machined workpiece was the lowest for a longer period of time. Under dry friction conditions, the surface of the Cu-Ni electrode-machined workpiece did not have any significant cracks compared with the surfaces of the untreated workpiece and the Cu electrode-machined workpiece. This contributed to the improved wear resistance of the machined surface.

Keywords Polycrystalline diamond · Electric discharge machining · Ultrasound · Graphene · Cobalt removal · Frictional wear · Powder · Powder-Mixed EDM

1 Introduction

Polycrystalline diamond is formed by single-crystal diamond and conductive binders sintered under high temperature and high pressure. In addition to high hardness, high wear resistance, good thermal conductivity, and low coefficient of friction, it also has the advantages of high grinding performance and good self-sharpening [1]. Currently, it is widely used in various fields, such as cutting tools, drilling tools, as well as automotive and aviation industries. Nevertheless, the high hardness of polycrystalline diamond material and the difficulty of isotropic material to fracture along a single disintegration surface [2] and other shortcomings make conventional machining methods not applicable to the processing of PCD.

Electrical discharge machining (EDM) is commonly employed in the machining of high-hardness, high-strength, and wear-resistant materials to remove material in a dielectric fluid through periodic pulse discharges between the electrode and the workpiece [3]. EDM is one of the common machining methods for PCD at present. Many researchers have conducted in-depth research on EDM PCD. Fernandez M.G et al. [4] conducted orthogonal experiments with different electrical parameters and obtained the optimal machining parameters by studying the signal-to-noise ratio of material removal rate and cutting speed. Ahmad M. A et al. [5]processed PCD with Cu electrodes under finishing conditions, revealing that the electrical parameters in finishing exerted a significant effect on the surface roughness and that the discharge current had no significant effect on the material removal rate. Rahim M. Z et al. [6] investigated the effect of process parameters on surface quality during the

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EDM grinding of PCD, suggesting that the high temperature generated during the discharge process brought about the transformation of the diamond phase below the surface into the graphite phase [5]. However, the EDM of PCD presents several challenges. Arcing may be generated in EDM, triggering transient temperature differences and short circuits, and thus deterioration of surface integrity. Moreover, since diamond particles are not electrically conductive during EDM, spark discharge occurs first at the conductive binder and is preferentially removed. This results in the explosive removal of diamond particles by the discharge energy, leading to deep craters and cracks.

This study aimed to address these issues. Wang P. X et al. [7] investigated the effect of high-frequency ultrasonic vibration on micro-electro discharge machining, revealing that the efficiency of ultrasound-assisted EDM can be improved by 82.7% and that the electrode loss can be reduced by 68.7% compared with that of ordinary EDM. Ou S. F et al. [8] discovered that adding hydroxyapatite powder to the medium during EDM machining of titanium alloys contributed to other surfaces and thinner recast layers. Chaudhari et al. [9] pioneered the introduction of MWCNT nanofluids into WEDM of Udimet 720, achieving significant improvements in surface integrity and machining efficiency through synergistic optimization of process parameters. Their subsequent comparative studies revealed that nano-graphene exhibited superior comprehensive performance over MWCNTs in PMEDM, accompanied by substantially enhanced machining efficiency and surface quality of Udimet-720 [10]. Vora et al. [11] reported that modifying the WEDM process with Expanded Graphite (EG) nano-powder considerably reinforced the machining efficiency and surface quality of Ti-6Al-4 V alloy. Wang X. Z et al. [12] designed a process of graphene mixed powder EDM for PCD machining. Their experimental studies demonstrate that the removal rate of PCD material processed by this method was many times higher than that of ordinary EDM and that the relative wear of the electrode was reduced by 70%. This enhancement stems from graphene's exceptionally high electrical conductivity, which enables uniform dispersion of discharge energy and prevents localized arcing, together with its high thermal conductivity that rapidly dissipates heat from the discharge zone, contributing to minimizing the heat-affected zone. Furthermore, its nanostructured flake morphology fills discharge gaps and optimizes the stability of the discharge channel [13]. These powder characteristics directly govern machining efficiency and surface integrity outcomes in EDM processes. Our research team [14] has previously explored ultrasound-assisted mixed-graphene powder electrical discharge machining (U-PMEDM) of PCD. The results specify that the synergistic effect of graphene and ultrasound significantly improved the material removal rate and the surface quality and that the method is of great significance for efficient and high-quality machining of PCD. During the EDM process, the powder formed a coating of intermetallic compounds and carbides on the machined surface, thus increasing the microhardness [15]. Furthermore, the incorporation of graphene powder led to surface modification of the PCD workpiece and thus phase transition [16]. This improved the microhardness and wear resistance of the PCD machined surface. Hence, the U-PMEDM method of machining PCD was employed in the present study to achieve a better surface machining quality.

Additionally, cobalt is the most commonly used metal binder in PCD, primarily serving to bond diamond particles together and form a stable polycrystalline structure. However, cobalt has a low melting point and hardness, is prone to softening or melting during the cutting process of the tool, and is subjected to oxidation or corrosion at high temperatures, in humidity, or in certain chemical environments. This will destroy the structural stability of the PCD and make the diamond particles fall off irregularly. Moreover, cobalt catalyzes the graphitization of diamond particles during PCD tool operation, compromising hardness and increasing wear susceptibility. As a result, the service life of PCD tools is shortened, and thus the cobalt content on the surface of PCD has a significant impact on its wear resistance. Qin W. B et al. [17] experimentally demonstrated that the wear resistance of PCD composite sheets was enhanced after the cobalt content was decreased. Huang H. F et al. [18]revealed that reducing the residual cobalt content in the PCD layer contributed significantly to reinforcing the wear resistance of PCD cutting teeth. In practice, PCD is generally decobaltized [19].

Vhareta M et al. [20]removed cobalt by corrosion solution leaching, and the cobalt removal did not damage the cohesion of the diamond substrate. Li T. J et al. [21] realized the selective removal of surface cobalt binder by pulsed laser ablation under appropriate parameters. When PCD is adopted to make tools, there is a need to both accurately remove the material through machining and to reduce the cobalt content to slow down wear in the application. While chemical removal of cobalt pollutes the environment and wastes energy, laser removal of cobalt creates a large heat-affected zone, making it difficult for existing machining methods to achieve both requirements simultaneously.

In the aforementioned studies, the use of mixed powders and ultrasonic assistance has brought about significant advancements in improving the removal rate and surface quality of PCD. Nonetheless, these studies have predominantly employed traditional electrodes such as copper while ignoring the effects of different electrode materials and the synergistic interaction with powders on the machining process. Furthermore, the effective cobalt removal from

PCD typically requires independent post-processing steps, such as acid leaching or laser ablation, resulting in complex process flows and potential issues of environmental pollution or thermal damage. Therefore, developing an integrated new process capable of simultaneously achieving efficient PCD machining and cobalt removal in a single operation holds substantial theoretical significance and practical value.

In summary, U-PMEDM has been proven to effectively improve the machining efficiency and surface quality of PCD, and surface cobalt removal has been established as a crucial step in enhancing its wear resistance. Nevertheless, existing technological approaches still have limitations. Specifically, machining and cobalt removal are two independent processes that fail to simultaneously tackle the issues of precise removal and environmentally efficient processing. The core innovation of this study lies in the new application of Cu-Ni electrodes in the U-PMEDM process for PCD. This forms a method that ingeniously utilizes the discharge reactions during EDM, allowing the Ni element to indirectly catalyze and promote the removal of cobalt from the PCD surface [22]. This study investigates the impact of different discharge energies on the machined surface quality and, more importantly, through a comparative analysis of the machining effects of Cu and Cu-Ni electrodes, elucidates the cobalt removal mechanism of the Cu-Ni electrode. The results verify the effectiveness of this integrated process in reducing the cobalt content on the PCD surface and improving its tribological properties.

2 Experiments

2.1 Experimental Preparation

In this experiment, a mirror EDM (Dongguan Rixin Co., Ltd., A30) was used to process the PCD workpiece. An ultrasonic vibration device was installed on the worktable, as illustrated in Fig. 1. The ultrasonic amplitude rod was connected to the workpiece through hexagonal bolts. Then, the ultrasonic device and the workpiece were fixed to the z-axis of the platform, so as to realize the vibration of the workpiece along the z-axis direction. During the machining process, the tool electrode was connected to the positive pole, the PCD workpiece was connected to the negative pole, and the machining process parameters were adjusted through the CNC system. The tool electrodes used in the study were Cu electrode and Cu-Ni electrode (Grade: B30 cupronickel, Nickel Content: 30.3%), with the specifications of Φ16mm×150 mm cylinders. The graphene powder was manufactured by Dazhan Nano (Guangdong) Co.

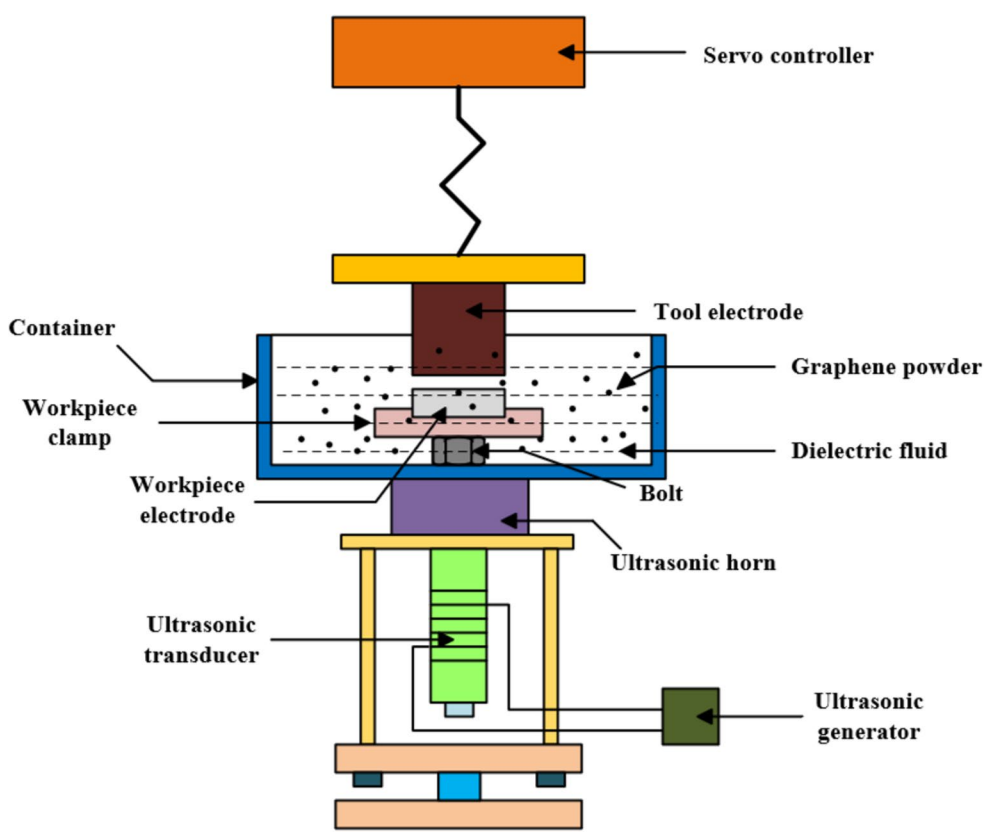


Fig. 1 Schematic diagram of the experimental setup

2.2 Physical characteristics of the workpiece

The workpiece used in this experiment was produced by Hunan Fengspeed Drilling Tools Co., Ltd., with a metal substrate thickness of 3.5 mm. The PCD layer consisted of a polycrystalline composite synthesized from single-crystal diamond particles with a crystal size of 6 μm and a conductive binder sintered under a high-temperature and high-pressure environment. Concerning the composition of this material, the diamond particles are about 85% by mass, and the binder is cobalt with about 15%. Graphene is difficult to adsorb on the surface of the workpiece because of the weak van der Waals interaction between graphene and the diamond surface [23]. The residual graphene powder on the surface of the workpiece can be effectively removed through ultrasonic cleaning. Following this characteristic, the addition of graphene powder as a component of the working fluid in the experimental process will not interfere with the measurement results of C elements on the workpiece surface.

2.3 Processing parameters and processing methods

In this study, a five-level discharge energy gradient experimental program was constructed. Cu electrodes and Cu-Ni electrodes were adopted to process at five parameter levels respectively, so as to systematically analyze the effects of different discharge energies and electrodes on the cobalt removal effect of PCD workpieces. In the experiments, the processing voltage was 120 V, the graphene powder content was 0.05 wt%, and the ultrasonic vibration frequency was 20 kHz. The specific experimental parameters are detailed in Table 1. Before processing, graphene was slowly added into the stirred EDM working medium and continuously stirred for about 10 min with a stirring bar to make it uniformly dispersed. A brand new PCD workpiece was utilized for each machining and was cleaned by an ultrasonic cleaner for 15 min before and after machining. According to the results of the preliminary experiments, a processing duration of 30 min can ensure sufficient material removal, stabilizing and homogenizing the surface condition of the sample. This duration satisfies the requirements for subsequent surface element analysis and avoids data deviations induced by insufficient machining, providing ample and stable reaction time for the discharge process to effectively remove cobalt elements. Therefore, a machining time of 30 min was set for each workpiece. The electrode rods were polished before each processing to ensure the reproducibility of the experiments. Moreover, each set of machining experiments was repeated three times to lower the uncertainty in the experiments.

Table 1 Experimental parameters

Parameter level	I	II	III	IV	V
ON (μs)	60	90	120	150	200
OFF (μs)	4	8	12	16	20
IP (A)	1.5	3	4.5	6	7.5

3 Results

3.1 Cobalt removal mechanism analysis

When a Cu-Ni electrode is adopted for EDM of polycrystalline diamond, the removal of cobalt is a complex process. Cobalt, as a binder for PCD, exerts a direct impact on the machining quality and the surface morphology of the workpiece by its removal mechanism during the EDM process. Regarding the processing mechanism, the removal mechanism of cobalt can be mainly attributed to the following two perspectives:

1) High temperature melting and oxidation
   The essence of EDM is to use the transient discharge between the electrode and the workpiece to generate high temperature, allowing the material to be locally melted and removed. Particularly, the temperature generated by the instantaneous discharge can reach 8,000 ~ 12,000°C, far exceeding the melting point of cobalt, and cobalt has good electrical conductivity. Hence, cobalt undergoes transient high-temperature melting during EDM. The incorporated graphene powder, attributed to its superior electrical conductivity, uniformly disperses discharge energy to generate numerous micro-scale and stable discharge channels. Concurrently, high-frequency ultrasonic vibrations perpendicular to the workpiece surface trigger intense cavitation effects and fluid flow dynamics, significantly enhancing the circulation efficiency of working fluid. This synergistic mechanism provides robust mechanical momentum for the detachment and transportation of molten cobalt from diamond particle interfaces, enabling efficient cobalt removal through fluidic forces. When the temperature of the local area is high enough, part of the cobalt will react with oxygen to generate cobalt oxide that is easy to detach from the PCD [24]. This further promotes the removal of cobalt under the combined effect of ultrasonic vibration assistance, discharge energy impact, and dielectric fluid, leaving a shallow and uniform small crater at the initial removal of cobalt under the effect of graphene powder. The mechanism is depicted in Fig. 2.
2) Indirect catalytic effect of nickel
   Cu-Ni electrodes have a higher melting point and better thermal stability and resistance to galvanic

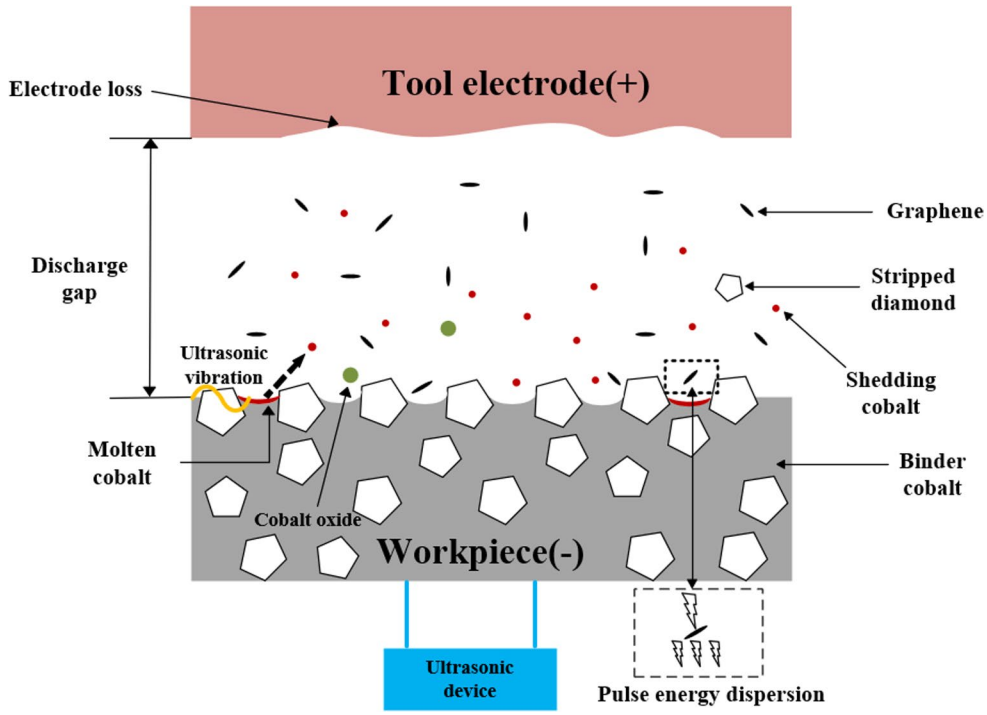


Fig. 2 High temperature melting and oxidation mechanism

corrosion compared to Cu electrodes [25], contributing to the improved stability of the process and allowing for more uniform cobalt removal. In EDM of PCD, the removal of diamond particles mainly relies on the discharge explosion force and graphitization effect. Moreover, Ni in the Cu-Ni electrode catalyzes diamond graphitization in a high-temperature environment [26], prompting the removal of diamond to rely more on the graphitization process. Graphite demonstrates excellent electrical conductivity and a loose structure that is prone to detachment. During the diamond graphitization removal process, thus, graphite flows toward the small pits where cobalt has been partially removed under the action of ultrasonic vibration perpendicular to the machined surface. This reinforces the electrical conductivity at the small pits and promotes the removal of residual cobalt. The directional driving force of ultrasonic vibrations during this process forms favorable conditions for subsequent sustained stable discharge. Hence, nickel can indirectly promote cobalt removal by catalyzing diamond graphitization, serving as one of the key mechanisms for cobalt removal from Cu-Ni electrodes during ultrasound-assisted mixed-powder EDM. With the gradual removal of cobalt, the initial small pits gradually expand and evolve into a more uniform pore structure. They may accommodate abrasive chips and grease during friction and have a positive effect on wear reduction. The mechanism is exhibited in Fig. 3.

3.2 Surface quality analysis

During EDM, the energy released per discharge increased as the electrical parameters rose, causing more material on the surface of the workpiece to be rapidly melted and removed in a short period of time. While this improved the material removal efficiency, it intensified the discharge impact effect. This can significantly influence the morphology, roughness, and microstructure of the machined surface.

The surface morphology of PCD workpieces machined by Cu electrode and Cu-Ni electrode under different electrical parameters is illustrated in Fig. 4. Among them, Figs. 4(a)-(e) present the surface morphology of PCD workpieces processed by Cu electrode. Figures 4(f)-(j) demonstrate the surface morphology of PCD workpieces processed by Cu-Ni electrode. The surface morphology of PCD processed by the Cu electrode was analyzed under different electrical parameters. The results suggest that at lower electrical parameter levels I, II, and III, the discharge energy was not enough to explosively and completely remove the processed fragments of cobalt and diamond particles. Moreover, the formed remelts were scattered on the surface, presenting a granular or speckled distribution. As the discharge energy increased, more binder and diamond were removed at the electrical parameter level IV. Most of the molten material was explosively removed, the amount of remelts decreased, and more diamond particles were exposed on the surface. Under electric parameter level V, the local temperature was very high during the discharge process because of the maximum discharge energy, bringing about the rupture

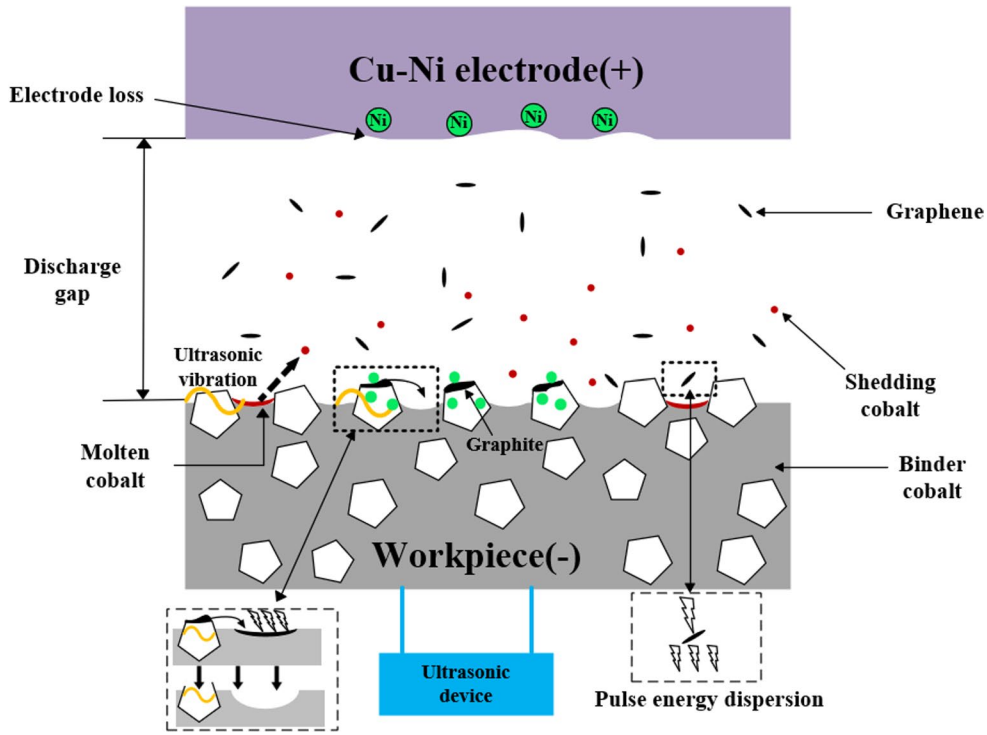


Fig. 3 Indirect catalytic effect of nickel mechanism

of diamond particles and the dramatic removal of binder. Consequently, large irregular discharge craters appeared on the surface.

As revealed by analyzing the morphology of the PCD surface processed by Cu-Ni electrode under different discharge parameters, the Cu-Ni electrode presented more surface pores and more exposed diamond particles than the Cu electrode under the same energy level. At discharge parameter levels I, II, and III, the lower discharge energy was insufficient to remove excessive binder material, resulting in fewer surface pores. Moreover, the lower discharge explosion force prevented efficient removal of the remelted material and its distribution over the pores. Discharge parameter level IV can produce the appropriate discharge energy. It can not only remove a large number of binders, increasing the surface pores, but also make the molten material be discharged smoothly, curtailing the accumulation of remelted material on the surface. When discharge parameter level V was employed for processing, too high a discharge energy produced a large ablation pit, and a large number of surface materials that rapidly melted could not be discharged in time. As a result, a considerable number of remelted materials accumulated on the surface, and then an uneven surface morphology appeared. Therefore, the surface quality was the best under the electric parameter level IV, the pore morphology was the best, and the distribution was the most uniform.

The surface roughness values (Ra) in this study were measured by a Beijing Times TR200 roughness tester. Measurements were obtained by the profile method, tracing a single path across the machined surface. The procedure adhered to the specifications of Chinese National Standards GB/T 3505−2009 and GB/T 10,610−2009. Five repeated measurements were conducted for each experimental condition, with the corresponding data presented in Table 2. The surface roughness of the workpieces machined by Cu-Ni electrode was greater than that of the workpieces machined by Cu electrode at all parameter levels. Under parameter level I, the surface roughness of the PCD workpiece processed by Cu-Ni electrode was 2.31 μm, which was 17% higher than that of the workpiece processed by Cu electrode. Meanwhile, the surface roughness of the Cu-Ni electrode processed by the Cu electrode was increased by 19% and 16% after processing at parameter levels IV and V, respectively. The surface roughness of the workpiece processed by the Cu-Ni electrode was higher than that of the workpiece processed by the Cu electrode. The cobalt removal mechanism analysis reveals that the nickel in the Cu-Ni electrode exerted an indirect catalytic effect on the removal of cobalt at high temperatures. At the same discharge energy, the Cu-Ni electrode removed more cobalt compared to the Cu electrode. Since a large number of pores and diamond bare phenomena were produced on the surface of the PCD when binder cobalt was removed in large quantities, the surface flatness was poor, and the roughness was larger.

As illustrated in Fig. 5, the surface roughness of PCD workpieces machined by Cu electrode and Cu-Ni electrode increased with the increasing EDM discharge energy. This

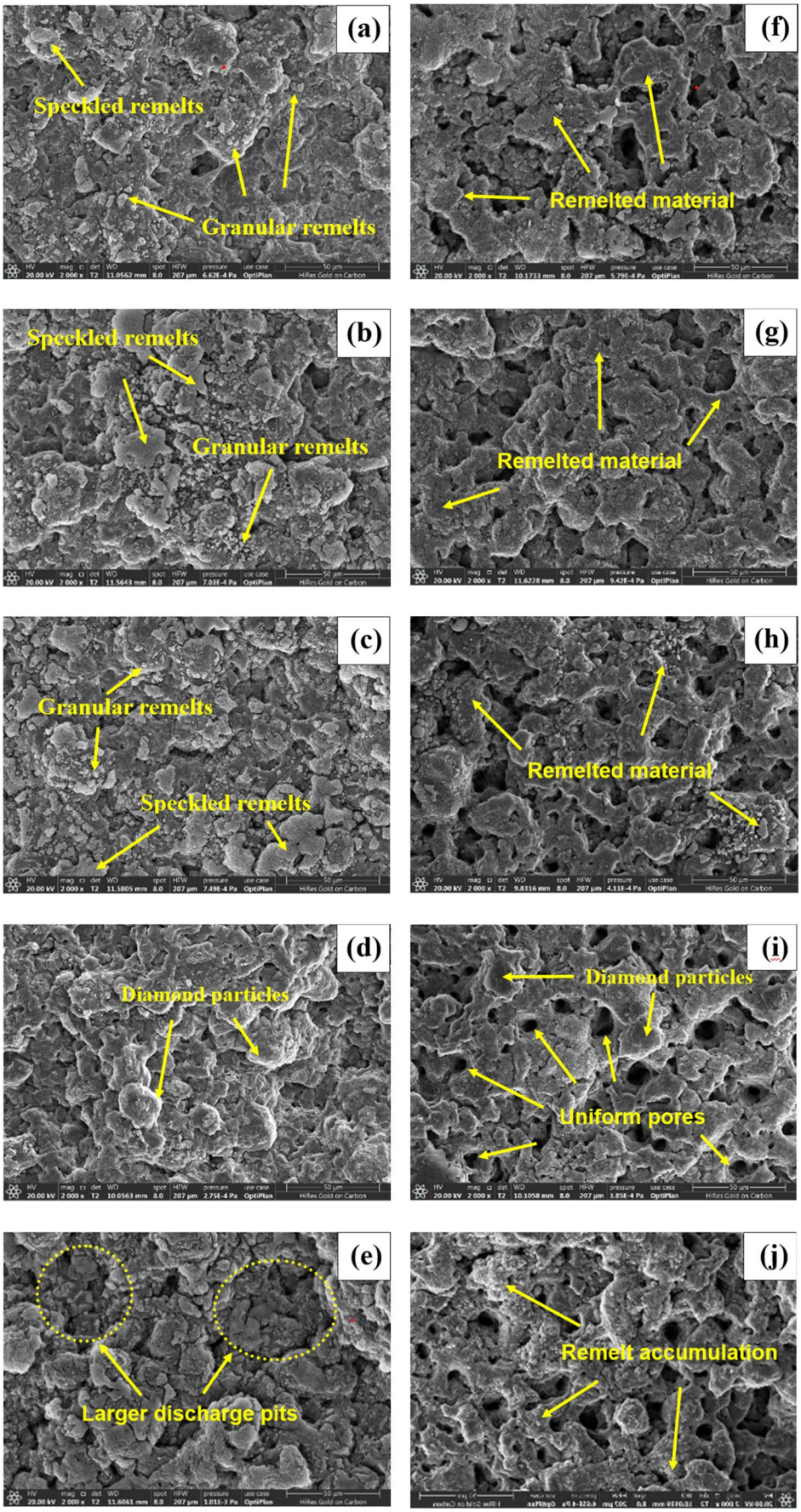


Fig. 4 The surface morphology of PCD workpieces machined with Cu and Cu-Ni electrodes at different parameter levels I-V; (**a**)-(**e**) Surface morphology after being machined with the Cu electrode; (**f**)-(**j**) Surface morphology after being machined with the Cu-Ni electrode

Table 2 The surface roughness values of PCD machined by two types of electrodes under different parameters

Run	Electrode	I	II	III	IV	V
		SR(μm)	SR(μm)	SR(μm)	SR(μm)	SR(μm)
1	Cu	2.01	2.13	2.29	2.59	2.96
2	Cu	2.06	2.01	2.3	2.57	2.9
3	Cu	1.96	2.11	2.39	2.75	3.12
4	Cu	1.90	2.22	2.32	2.71	3.05
5	Cu	1.97	2.08	2.45	2.63	3.02
Average	Cu	1.98	2.11	2.35	2.65	3.01
Standard Deviation	Cu	0.0598	0.07649	0.06819	0.07746	0.08426
1	Cu-Ni	2.27	2.57	2.81	3.17	3.39
2	Cu-Ni	2.39	2.51	2.73	3.13	3.49
3	Cu-Ni	2.22	2.54	2.88	3.12	3.46
4	Cu-Ni	2.33	2.62	2.82	3.11	3.52
5	Cu-Ni	2.34	2.51	2.76	3.22	3.59
Average	Cu-Ni	2.31	2.55	2.80	3.15	3.49
Standard Deviation	Cu-Ni	0.06596	0.04637	0.05788	0.04528	0.07382

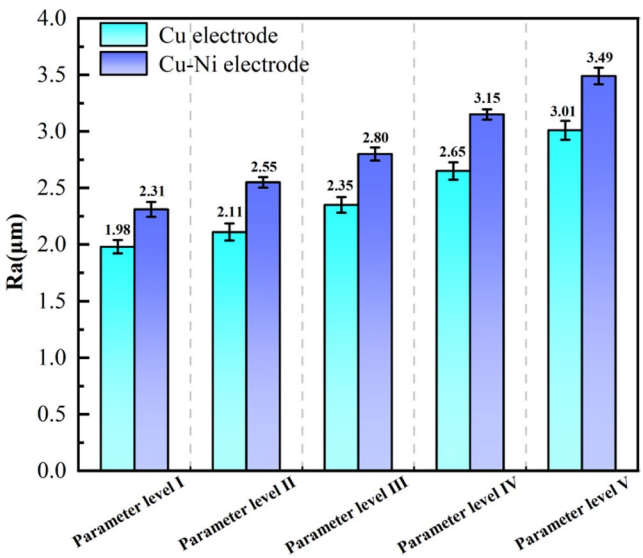


Fig. 5 Surface roughness of PCD machined by two types of electrodes under different processing parameters. Data are presented as mean ± standard deviation (SD)

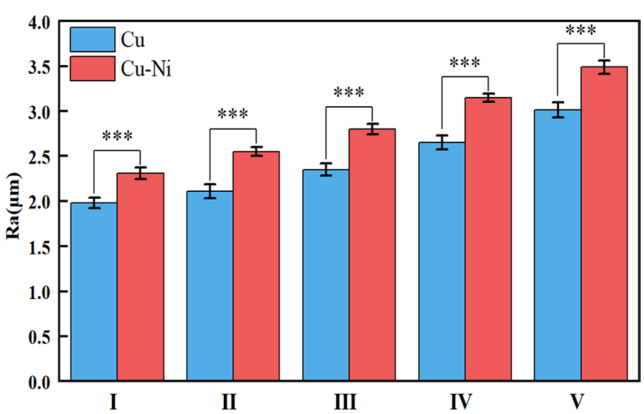


Fig. 6 Statistical analysis of surface roughness of PCD machined by two types of electrodes under different parameters (t-test)

was because the thermal effect of the EDM on the PCD surface material was reinforced as the discharge energy increased, resulting in larger surface discharge ablation marks, uneven surface, and increased surface roughness value.

As suggested in Fig. 6, an independent samples t-test was conducted to analyze the difference in surface roughness (Ra) of PCD machined with the two types of electrodes. The results demonstrated highly statistically significant differences in the mean Ra values between the Cu electrode and the Cu-Ni electrode under all parameter sets ($p < 0.001$). Combined with the data from Table 2, the small standard deviations observed for both groups indicate good experimental repeatability. These findings specify that differences in surface roughness between the two electrodes stem from the intrinsic properties of the electrode materials, which fundamentally influence the surface quality of the machined PCD, rather than from random variation.

3.3 Analysis of element distribution on different workpiece surfaces

Parameter level IV achieved an optimal balance in discharge energy. It provided sufficient energy to effectively remove the binder and create a considerable number of uniformly distributed pores while avoiding the issue of excessive accumulation of remelted material induced by overly high energy levels. This uniform pore structure ensured effective heat dissipation and accommodated wear debris during the subsequent friction process of the workpiece, thereby mitigating wear. Furthermore, the smallest standard deviation of the surface roughness of PCD machined with the Cu-Ni electrode was obtained under parameter level IV, as evidenced by the data in Table 2. This indicates that the machining process with the Cu-Ni electrode achieved the

Table 3 The Cobalt content on the surface of PCD machined by different electrodes

Run	Untreated (%)	Cu electrode (%)	Cu-Ni electrode (%)
1	11.20	6.96	4.10
2	11.25	7.00	4.20
3	11.33	7.06	4.25
4	11.40	7.10	4.30
5	11.47	7.18	4.35
Average	11.33	7.06	4.24
Standard Deviation	0.10932	0.08602	0.09618

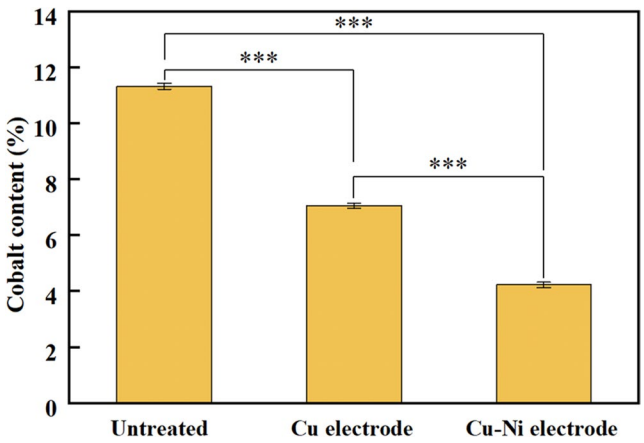


Fig. 7 Analysis of cobalt content across the untreated and differently electrode-processed workpieces (One-way ANOVA). Data are presented as mean ± standard deviation (SD)

best stability and the highest repeatability under these specific conditions. Thus, Cu electrode and Cu-Ni electrode in parameter level IV machined PCD workpieces and untreated PCD workpieces were selected for comparative analysis of surface elements. Additionally, the cobalt content on the PCD surface was specifically investigated. With the purpose of guaranteeing the representativeness of the analysis, five distinct areas were selected for measurement, covering both the central and edge regions of the sample. The corresponding statistical data are summarized in Table 3. A one-way analysis of variance (ANOVA) is illustrated in Fig. 7.

Figure 8 displays the distribution of different elements on the surface of the unprocessed workpiece. In the unprocessed PCD workpiece, Co, acting as a binder, was generally distributed between the diamond particles in a spherical or island shape, along the diamond grain boundaries. Simultaneously, it played the role of binding the diamond particles together. The boundary between cobalt and diamond was insignificant, and the diamond particles were wrapped by the cobalt binder. Therefore, the removal of cobalt exerted a significant effect on the surface morphology and structure of diamond. After cobalt removal, diamond particles were exposed on the surface of the workpiece, and the boundary between particles became more significant.

Figure 9 exhibits the distribution of different elements on the surface machined by the Cu electrode. After the workpiece was processed by Cu electrode, the distribution of Co element was more dispersed and covered the surface

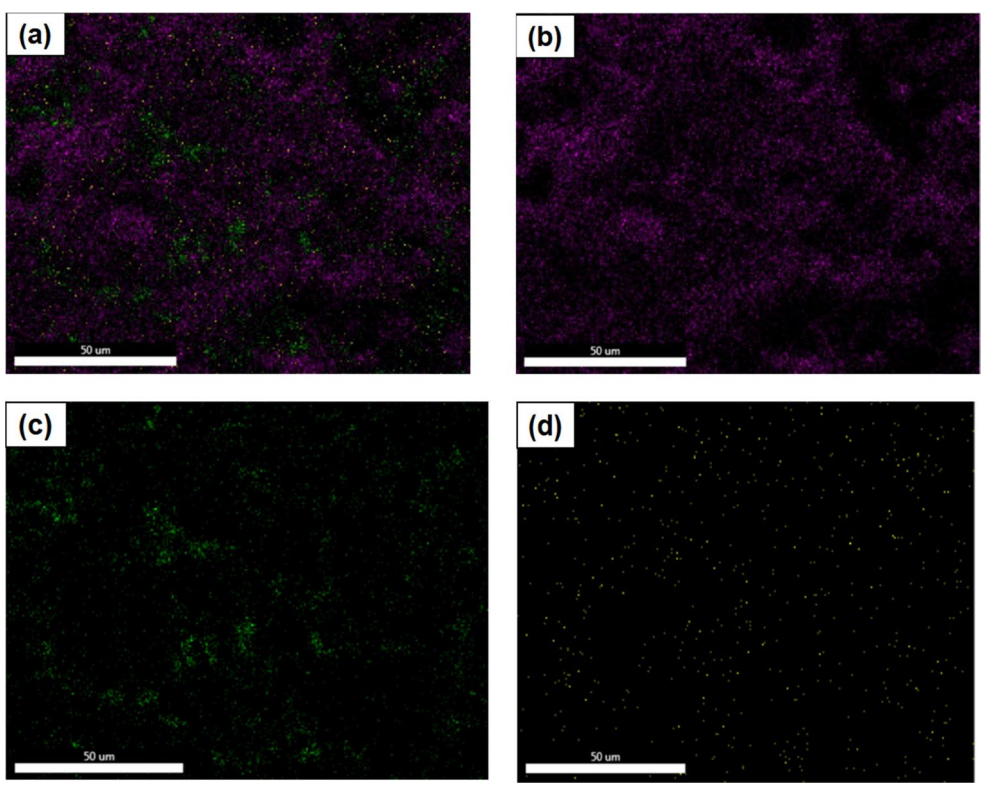


Fig. 8 Distribution of different elements of the untreated workpiece. (**a**) Total elemental distribution; (**b**) C elemental distribution; (**c**) Co elemental distribution; (**d**) Ni elemental distribution

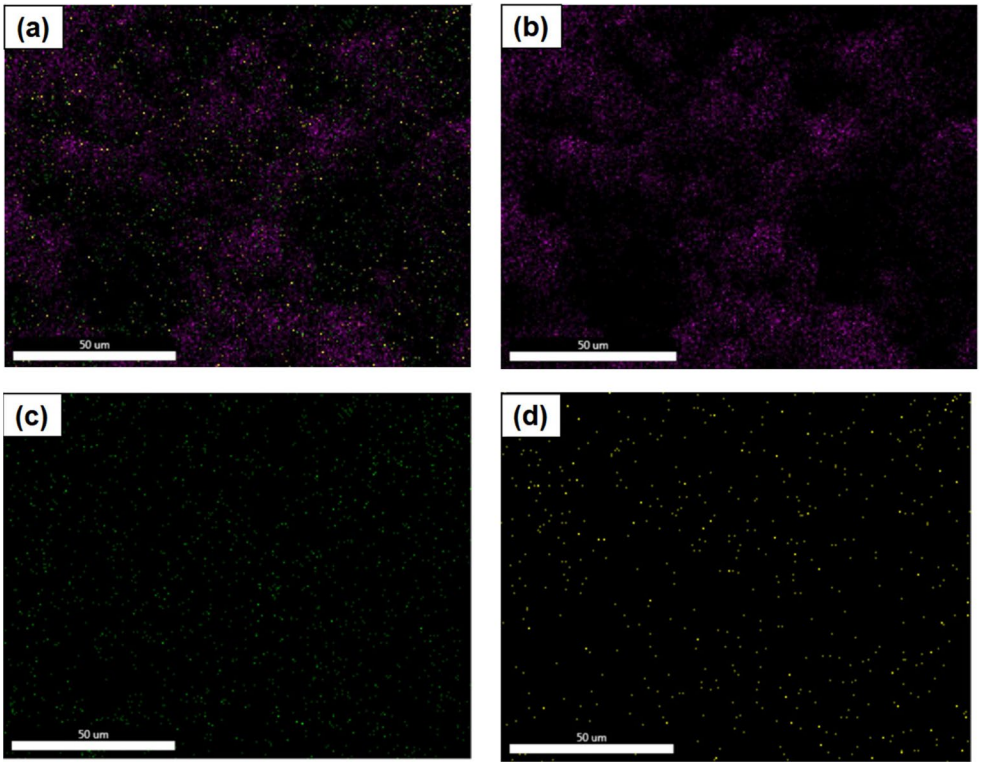


Fig. 9 Different elemental distributions of workpieces machined by Cu electrodes. (**a**) Total elemental distribution; (**b**) C elemental distribution; (**c**) Co elemental distribution; (**d**) Ni elemental distribution

of diamond particles composed of C element. The surface cobalt content analysis reveals that the Cu electrode-machined PCD workpiece exhibited an average cobalt content of 7.06%, which was about 38% lower than that of the unprocessed workpiece (Table 3). This reflects that spark discharge with a Cu electrode reduced the cobalt content, but did not achieve significant cobalt removal. Specifically, the cobalt reduction primarily stemmed from the good electrical conductivity and low melting point of cobalt being removed by the high temperature generated by the discharge.

Figure 10 displays the distribution of different elements on the surface of the workpiece processed with a Cu-Ni electrode. On the surface of the PCD workpiece processed by Cu-Ni electrode, the Co element was mainly distributed in the junction of diamond particles, hardly covered on the surface of the C element. Moreover, the boundary between the C element was significant, and the diamond particles composed of it were considerably exposed. As demonstrated in Table 3, the Cu-Ni electrode-machined PCD surface exhibited an average cobalt content of 4.24%, which was 63% lower than that of the unprocessed workpiece, accompanied by a significant removal effect. The ANOVA results in Fig. 7 reflect that the cobalt content on the PCD surface machined with the Cu-Ni electrode was significantly lower than that achieved with the Cu electrode. The statistical test yielded a result of $p<0.0001$, revealing that the difference was highly statistically significant. Therefore, the machining performance of the Cu-Ni electrode was indeed superior to that of the Cu electrode. Cu-Ni electrode exerted a catalytic effect on the material of PCD through the indirect catalytic effect of nickel. Thus, during diamond graphitization, graphite fell into the cobalt under the action of ultrasonic vibration. Initially, the cobalt discharge high temperature melts to remove the small pit, contributing to promoting cobalt discharge removal. This is an important reason why the cobalt content of workpieces machined by Cu-Ni electrodes is lower than that of workpieces machined by Cu electrodes. Second, the Cu-Ni electrode has a high melting point and better resistance to galvanic corrosion, making it less prone to high-temperature depletion than the Cu electrode. Additionally, its discharge characteristics may stabilize the high-temperature effects in the machining area, so as to improve the cobalt removal efficiency. Attributed to the synergistic effect of these mechanisms, the Cu-Ni electrode EDM of PCD was more effective in removing cobalt, enabling the exposure of more diamond particles and the improvement of the tribological properties of PCD.

4 Tribological properties

The tribological properties of Cu-Ni electrode-processed workpieces were further investigated to verify the effectiveness of this method for cobalt removal in practical applications. The workpieces processed under the parameter level

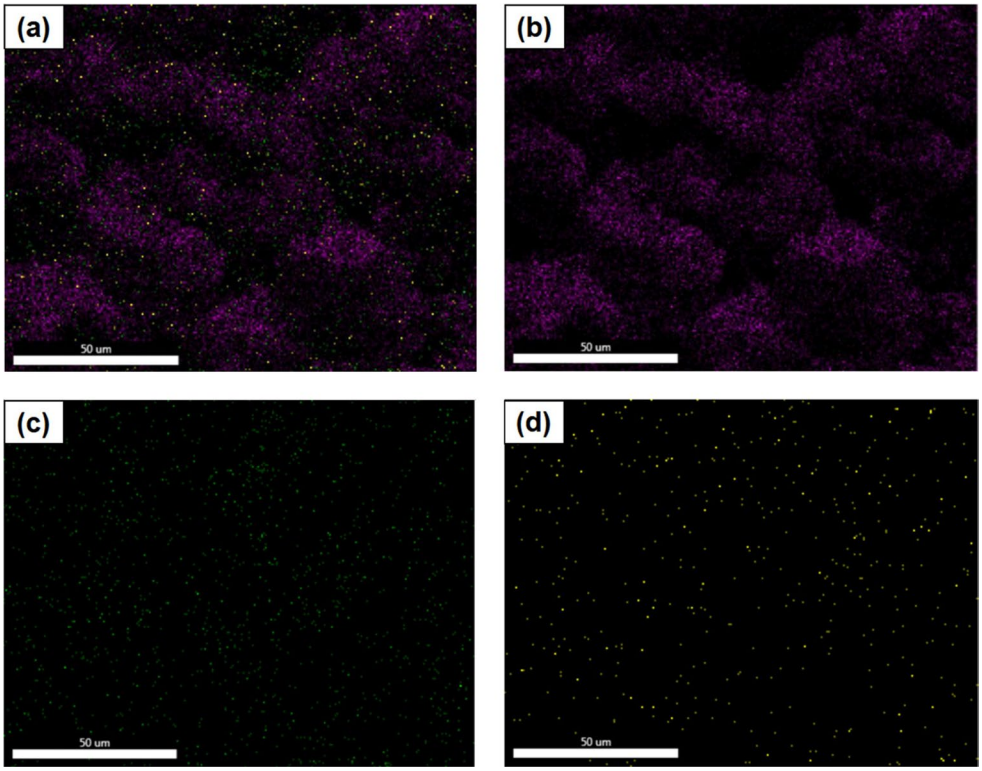


Fig. 10 Different elemental distributions of workpieces machined by CuNi electrode. (**a**) Total elemental distribution; (**b**) C elemental distribution; (**c**) Co elemental distribution; (**d**) Ni elemental distribution

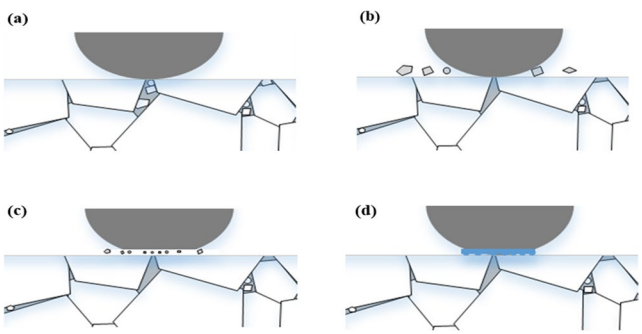


Fig. 11 Wear mechanism diagram during dry friction (**a**) crack generation stage, (**b**) particle shedding stage, (**c**) particle grinding stage, and (**d**) transfer film formation stage

IV with the best surface quality and the lowest cobalt content were selected. Six sets of dry and lubricated friction experiments were performed on the untreated workpieces and the workpieces processed by Cu and Cu-Ni electrodes, respectively. Higher hardness Si_3N_4 ceramic balls were selected as the counter-abrasive balls. Si_3N_4 ceramics, with excellent wear resistance, low coefficient of friction, and high compressive strength, maintain their shape and properties more stably when they are rubbed against the PCD workpieces, avoiding excessive wear due to abrasion.

Dry friction experiment is PCD workpiece and Si_3N_4 ceramic on the grinding ball friction surface between the friction surface without adding any lubricating medium, only through the workpiece surface of the reciprocating friction by direct contact between the workpiece surface; lubrication friction experiments in the friction surface with the addition of lubricating grease. MoS_2 was chosen as the lubricating grease to maintain a stable lubrication effect under high-temperature and high-pressure conditions. Moreover, it can form a thin lubricating film on the friction surface to effectively lower the occurrence of wear.

Figure 11 illustrates the dry friction mechanism of an untreated PCD workpiece with Si_3N_4 ceramic balls. During the friction process, the difference in thermal expansion coefficient between diamond and cobalt led to the stress concentration on the PCD surface, the reciprocating friction produced microcracks, and the nearby fine diamond particles took the lead in shedding [27]. Furthermore, the shedding particles were subjected to a chemical reaction on the friction surface. It transformed the particles from diamond to disordered graphite, amorphous carbon up to pure graphite. Moreover, it destroyed the D-D bond of diamond to trigger graphitization. Cobalt promotes graphitization in the pre-friction stage, and the graphite formed can be used as a solid lubricant [28], assisting in forming a transfer film on the friction surface and thus weakening the effect of friction during the friction process. However, prolonged frictional heat damages the transfer film [29], loss of cobalt oxidation leads to decreased diamond bonding and particle detachment, and microcracking is exacerbated by differences in thermal expansion, bringing about the significantly lowered wear resistance [30]. Given the low cobalt content

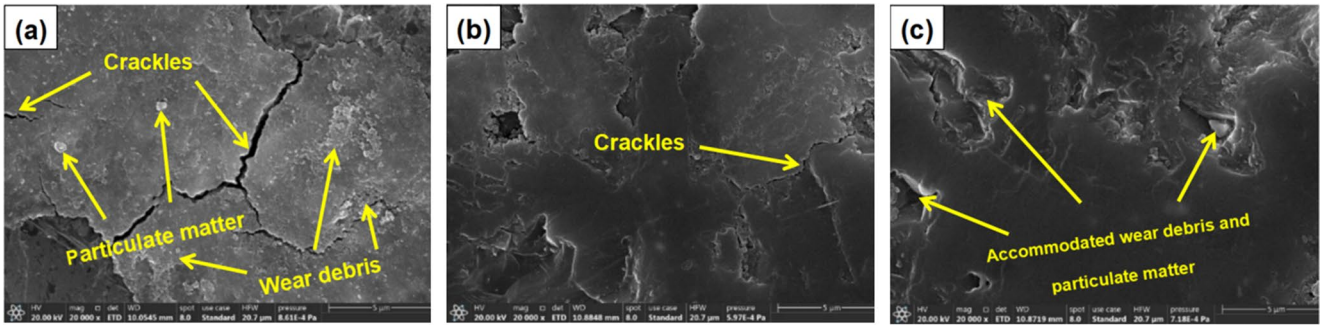
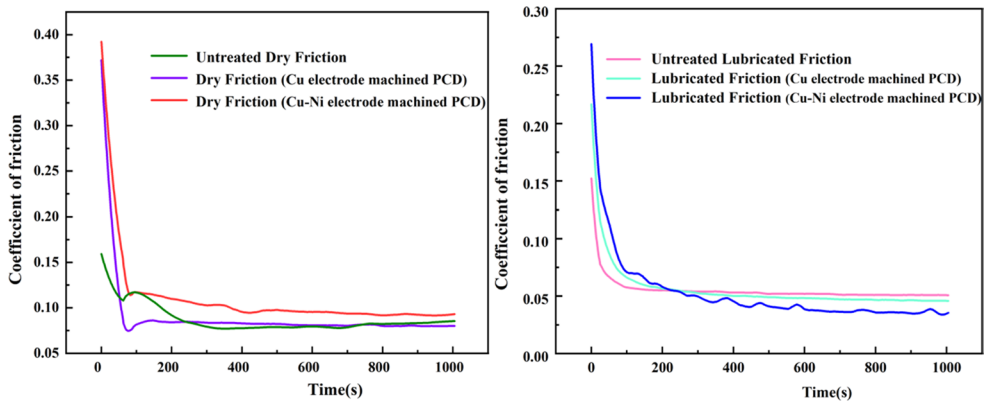


Fig. 12 Surface morphology of different workpieces after dry friction. (**a**) Dry friction surface morphology of untreated workpieces; (**b**) Dry friction surface morphology of workpieces processed with Cu electrode; (**c**) Dry friction surface morphology of workpieces processed with Cu-Ni electrode

Fig. 13 Friction coefficient curve



on the surface of the Cu-Ni electrode and the uniformity of the pores, cracks caused by differences in thermal expansion can be avoided, and diamond bonding can be stabilized. Moreover, the pores can hold the abrasive chips and lessen the abrasive wear and adhesive wear, thereby considerably improving the dry friction wear resistance. Similarly, in lubrication friction, the uniform pores retained grease to form a stable lubrication layer, and the physical and chemical reaction between the Si_3N_4 ceramic surface and MoS_2 generated a solid lubrication film, further slowing down the wear.

In Fig. 12a, the surface of the untreated workpieces has a high cobalt content, which generates thermal stresses under the difference in the coefficients of thermal expansion of the diamond. Concerning the effect of friction, this results in the appearance of cracks on the surface, especially expanding at the boundaries of the grains, and destroying the structure of the diamond [31]. Concurrently, in the cobalt-rich area, under the frictional heat and catalysis of cobalt, the boundaries of diamond grains are graphitized, and the particles are loosely peeled off. This reduced the surface finish, hardness, and wear resistance of the workpiece while leading to dry friction adhesion, scratches, and other issues [32]. In Fig. b, cracks, exposed particles, and spalling pits were produced on the surface of the workpiece. They originated from the frictional heat generated by the cobalt expansion to produce localized stress, weakening the diamond bonding force. The remelts generated by the Cu electrode machining were dislodged and adhered. Together with the dislodged diamond particles, they induced scratches and weakened the abrasion resistance and anti-cracking property. In Fig. c, the surface of the workpiece with low cobalt content was free of cracks because the low cobalt lowered the volume expansion, cracks, and diamond graphitization, and maintained the particle bonding force and hardness. The workpiece processed with a Cu-Ni electrode possessed uniformly distributed pores, which not only dissipated heat to avoid thermal damage but also accommodated abrasive chips to reduce abrasive wear. Moreover, the surface was smooth and flat after friction, forming a significant performance advantage when being used for processing superhard materials.

As revealed in Fig. 13, in the dry friction experiment, the PCD workpiece processed by Cu-Ni electrode exhibited the highest friction coefficient owing to the many surface pores and unevenness. Besides, the friction coefficient of the untreated workpiece was briefly lower because of the cobalt-catalyzed diamond graphitization at the initial stage. It was more than the friction coefficient of the workpiece processed by the Cu electrode, attributed to the accumulation of heat after 720s. During lubrication

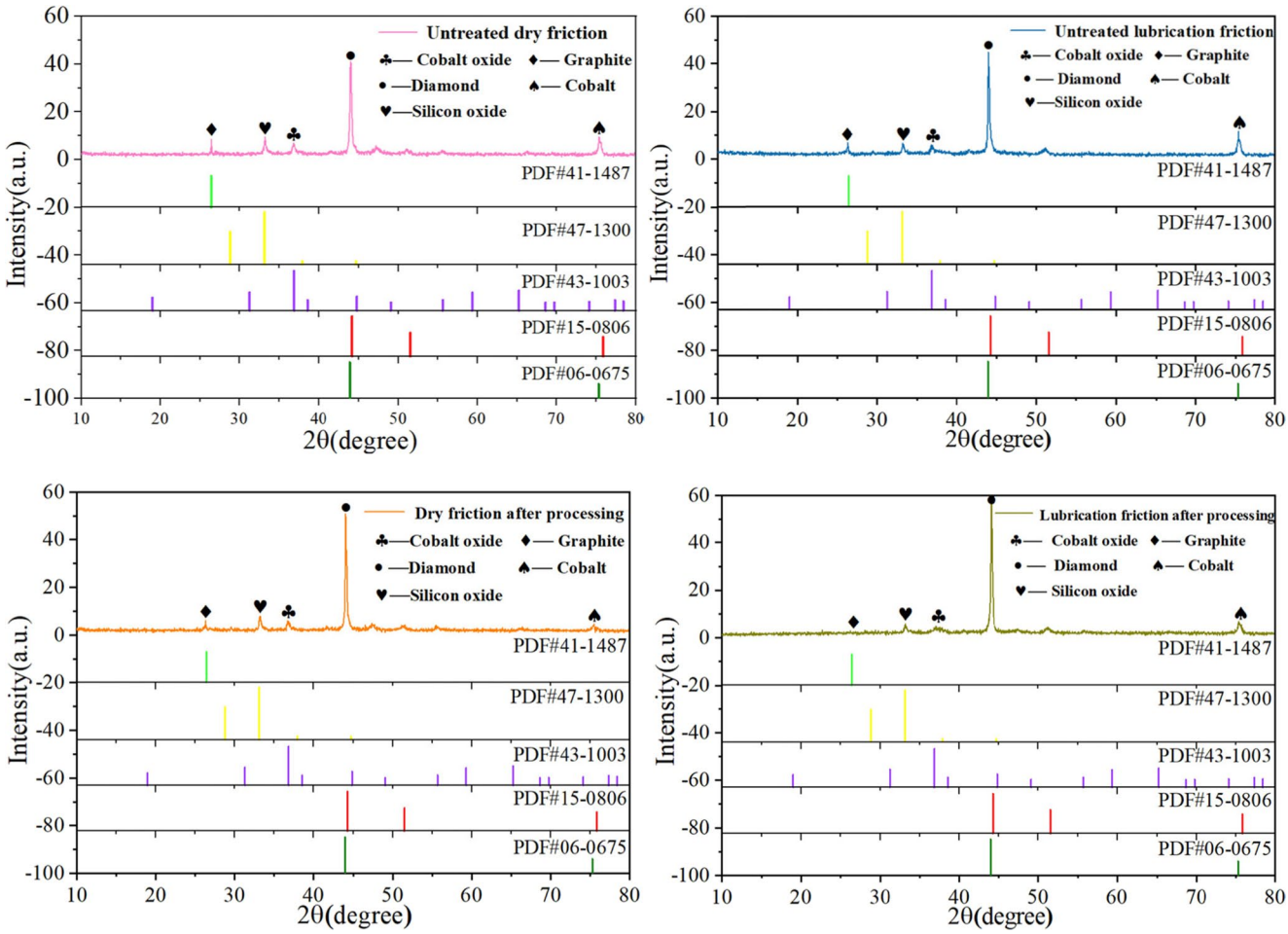


Fig. 14 Phase diagrams of untreated PCD and PCD machined by Cu-Ni electrode after friction under different conditions

friction, the overall friction coefficient was lower than that of dry friction, and the lowest friction coefficient appeared in the untreated workpiece within 100–250 s under the smooth surface. The lowest friction coefficient during long-term lubrication was observed in the Cu–Ni electrode-machined workpiece after 280 s, attributed to its surface porosity, reduced cobalt content, and the formation of a stable lubrication film.

Under the same friction conditions, the cobalt phase, graphite phase, silicon oxide phase, and cobalt oxide phase of the Cu-Ni electrode processed workpieces were weaker, and the diamond phase was stronger compared with the untreated workpieces, as suggested in Fig. 14. The Cu-Ni electrode processing optimized the composition of the workpieces' phases by removing the cobalt, reducing the oxidation reaction and the graphitization of the diamonds, enhancing the heat dissipation, and utilizing the pore structure to accommodate the debris and reduce the wear and tear. Consequently, the diamond phase was significantly enhanced, while other phases that were detrimental to the performance of the workpiece were weakened.

The proposed wear mechanisms are strongly supported by the results of elemental and phase analyses. For the untreated workpiece, EDS analysis (Fig. 8) specifies that the Co element tended to enrich at the grain boundaries, providing initiation sites for micro-crack propagation along these boundaries (Fig. 12a). Furthermore, XRD phase analysis (Fig. 14) confirms the significant presence of Co phase and graphite phase on the surface of the untreated workpiece, verifying the mechanism of cobalt-catalyzed graphitization of diamond during the friction process. In contrast, the wear performance of the workpiece processed by the Cu-Ni electrode was fundamentally improved. Quantitative EDS results (Table 3) unveil that the average Co content on its surface was reduced to 4.24 wt%, which mitigated the risks associated with thermal expansion and catalytic graphitization. Moreover, XRD results (Fig. 14) demonstrate a significant enhancement in the diffraction intensity of the diamond phase, as well as the markedly weakened diffraction intensities of detrimental phases such as Co and graphite. This evidence, at the phase level, confirms that the

Cu-Ni electrode effectively lessened the cobalt content and facilitated the formation of a uniform pore structure capable of accommodating wear debris (Fig. 12c). This not only suppressed the negative effects of cobalt but also stabilized the diamond structure, contributing to significantly optimizing the wear resistance of the workpiece.

5 Conclusion

This study confirms that in U-PMEDM of PCD, the adoption of Cu-Ni composite electrodes enabled an innovative "machining-and-decobaltification" integrated solution. Compared with traditional Cu electrodes, this approach significantly lessened the cobalt content on the PCD surface through the catalytic action of the Cu-Ni electrode while forming a uniform porous structure capable of accommodating wear debris. This represents not only a modification of surface morphology but also a fundamental enhancement of the material's subsurface properties. These improvements are evidenced by a significantly reduced friction coefficient under lubricated conditions and the integrity of the wear morphology, thereby considerably increasing the wear resistance and prolonging the service life of the PCD workpieces.

The practical significance of this technology lies in providing a new pathway for the efficient and precision manufacturing of critical, high-wear-resistant, long-life components such as PCD tools and molds. It holds the potential to eliminate or simplify subsequent independent processing steps, such as acid leaching for cobalt removal, which pose environmental concerns. Consequently, this method enhances processing efficiency while weakening environmental impact.

Nonetheless, this study has certain limitations. For instance, the universality of this process for PCD materials with different cobalt contents requires further verification. Moreover, the synergistic optimization between the Cu-Ni electrode and discharge parameters has not been explored. This direction could be considered in future research to further enhance the overall performance of the process.

Author Contribution All authors contributed to the study conception and design. Liujie Qi: Writing -original draft, Validation, Investigation, Data curation, Conceptualization. Li Li: Writing - review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition. Huiming Zhang: Methodology, Funding acquisition, Data curation. Tianyu Wang: Methodology, Data curation. Pengcheng Yu: Investigation. Bing Ma: Methodology. The first draft of the manuscript was written by LiuJie Qi and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Declarations

Conflict of interest The authors declare no competing interests.

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